

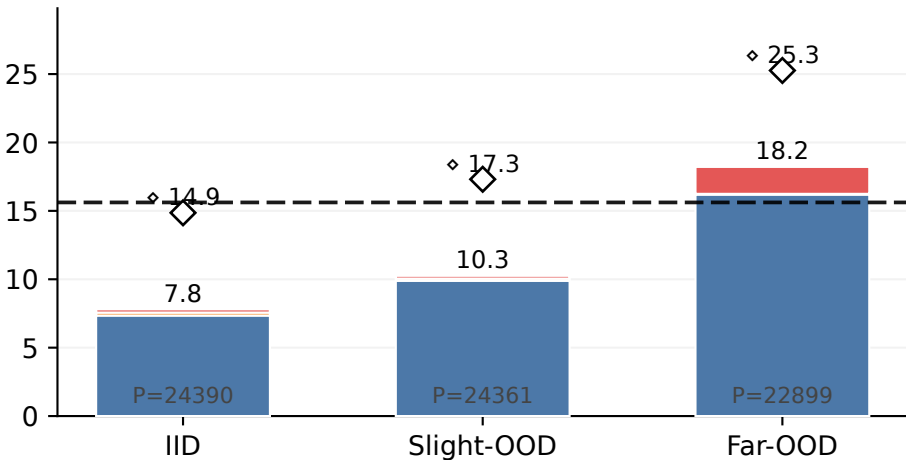
Generalization-Bound Decomposition (Top-K=64)

Empirical components shown as stacks; full bound marked with $\diamond$

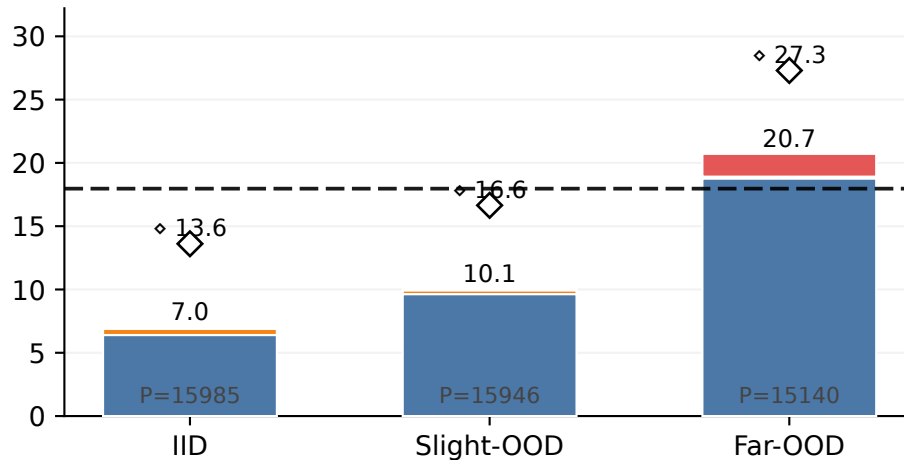
Empirical Risk (R) Reconstruction Gap (ϵ) Pool Mismatch (η_B) $\diamond$ Full bound - - Random baseline

Bits per Dimension (BPD)

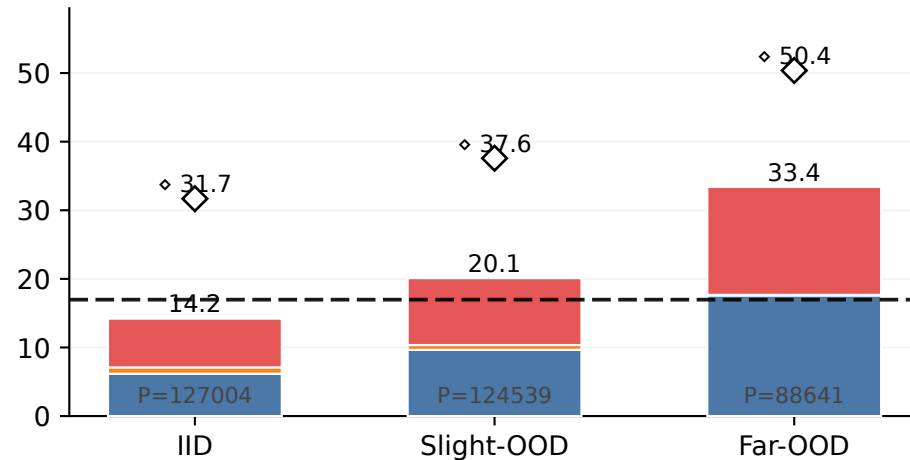
GPT-2 Small



Gemma-2B



Llama-3.1-8B



Data distribution